

Motivations to read and learn in videogame lore: the case of League of Legends

Marçal Mora Cantallops
University of Alcalá
Plaza de San Diego, s/n
Alcalá de Henares, Spain
marcal.mora@edu.uah.es

Miguel-Angel Sicilia
University of Alcalá
Plaza de San Diego, s/n
Alcalá de Henares, Spain
msicilia@uah.es

ABSTRACT

League of Legends is a multiplayer online battle arena game that follows a freemium model, but where the in-game transactions do little to impact a player's performance or ability. Although its characters, called Champions, can be purchased with actual (or in-game) money, there is a weekly rotation of ten free champions where players can test new champions before buying them. When new content is launched or a Champion is changed or reworked, players react to that by either using that Champion more or less (depending on whether they perceive it as positive or negative). Additionally, from time to time, Special Events are launched. The Bilgewater event, "Burning Tides", is one of such events, the only one that happened in season five, that aimed to merge lore with gameplay and represented Riot most ambitious story-telling experiment to date. Moreover, the seven Champions that had its background tied to Bilgewater were added to the free rotation on top of the other ten during the event, so there was no economic limiting factor that could hinder players from using them. This paper aims to understand how an event such as Bilgewater impact players in two dimensions; first, whether special events have an impact in Champion Usage and second, whether lore events can have an impact in player's motivation to read and learn in League of Legends. Results show that the effect in Champion Usage is unclear, with other factors being more relevant than Special Events, but that lore events have the potential to drive motivation to read and learn about the Champions and their story, events or optimal builds in the game, with a clear peak of interest in them.

CCS Concepts

• Information systems → Massively multiplayer online games • Applied computing → Computer games. • Theory of computation → Convergence and learning in games.

Keywords

League of Legends; Multiplayer Online Battle Arena; champion usage; Bilgewater; literacy

1. INTRODUCTION

Some 205 million people watched or played eSports in 2014 [1]. Among them, multiplayer online battle arena (MOBA) games are in the lead, both in players and watchers. In a few words, MOBA games are a subgenre of real-time strategy games in which two teams, typically consisting of five players each, compete against each other with each player controlling a single character. Competition takes place in a map with three lanes that connect one base (called nexus) to the opposite one. The main goal is to destroy the opposing nexus. Contrary to real-time strategy games, there is no unit or building construction in a MOBA game, so much of the strategy revolves around individual character development and cooperative team play in combat [2].

Two of the three most played PC games are MOBA games: League of Legends and DOTA 2 [3]. League of Legends in particular, developed and managed by Riot Games, has been leading the most played online game since 2012. Riot Games uses a Free-to-Play model for League of Legends; this means that the game can be downloaded and played for free but some content needs to be paid for, using either real currency (which is previously used to purchase "Riot Points") or in-game currency (in this case, named "Influence Points") which is obtained in various amounts at the end of each match. The main element of the game are the Champions, which are the player-controlled characters in League of Legends. Although there are a few other game modes, the main one (and the most popular) is Summoner's Rift, where five players per side chose one of these Champions each and compete against other team of similar skill, according to Riot's matchmaking algorithm. Two submodes can be identified here: Normal and Ranked. The main difference among both is that Ranked can be identified as the professional side of the game, where each match outcome counts towards an overall rank in the game, following the "League System", going from Bronze V to Challenger. Professional players are recruited in Challenger, that contains less than 0,1% of the Ranked player base, while most players play in Bronze, Silver and Gold, the bottom three categories, with more than 90% of the player base [5]. As outcome in Ranked matches matters, players usually choose the Champions they are more skilled with. Thus, to train on other Champions, Normal or other modes are preferred.

As of May 2015, there are 131 champions available, each with a different role (e.g. Assassin, Support, Tank), a different set of abilities (each Champion has, at least, four special abilities plus a passive) and different base stats. Therefore, and also heavily depending on the metagame [4] at the moment of play, every Champion is arguably better suited for a particular playing position. Over the years, these positions have evolved and are currently set according to the lanes: one "top", one "mid" and one "bot", with a "jungler" gathering the resources in the jungle between lanes and a "support" which would, in theory, have some degree of freedom to provide utility to the team but it is usually linked to the "bot" player/position. During the game, each player earns gold from

multiple sources and can use it to purchase multiple items that have the power to enhance the Champion. The combination of a Champion and the chosen items is called a “build”.

To introduce new features in the game while keeping the balance, Riot Games releases compulsory patches, usually biweekly. Besides new features, these patches are used to multiple ends, but in most cases they target Champions or items that became too dominant (thus being played by most of the players) or that are too weak in comparison (thus, nobody is playing them). Of particular interest are “reworks”, which happen whenever a Champion feels too out of date to Riot Games and it’s fully reworked with different looks and abilities.

Players don’t need to purchase Champions in order to play. The Free-to-Play model used by Riot Games provides a free Champion rotation consisting of ten different Champions every week. However, they can’t be used in Ranked. Free rotation Champions also provide a chance to try Champions that are not owned by the player, regardless of the intention of the player to purchase the Champion at a later stage or not. Champion cost varies; from some that are relatively non-expensive to the newest ones, which can be up to fourteen times more expensive (ranging from 450 influence points cost to 6300). Differences in usage over the free rotation Champions can give an idea of the popularity of each particular Champion.

Aside from Champions, the other main purchase element are skins. Skins are, to put it simple, appearances that can be purchased to customize a particular Champion. In any case, it’s important to note that skins have no impact in the basic gameplay; they are just visual candy, gimmicks that change the looks of the player-controlled character and/or his abilities.

Last, and usually relegated to a second plane, every Champion (and the game itself) has a background with some lore associated to it. “Back in the day, League of Legends used to offer relatively robust narrative explanations for why all the champions are dropping into the map and duking it out with one another over, and over, and over again. More recently, League developer Riot has promised to do a full overhaul of its game’s lore. Whatever the status of that narrative reworking may be, it hasn’t stopped Riot from churning out new content for League” [6]. While a complete lore is not currently available, some Champions share a theme that places them in the same story. When a new Champion is released, it is usually associated with one of these storylines although, in most cases, this in-game lore is either short or inconsequential. However, Riot Games launches, from time to time, events that temporarily change the game mode options to give players more variety (and thus more reason to keep logging on). Some of these events aim to merge lore with gameplay. The Bilgewater event, “Burning Tides”, is one of such events, and represented Riot most ambitious story-telling experiment to date.

2. DATA COLLECTION

Lee and Ramler (2015) analyzed the lasting impact the free rotation feature and new and updated content (such as new and updated champions, new skins and official game updates) have on champion usage. To do so, they developed a Python script that scrapped the data from the LoLDB Gameguyz website and decided to substitute its “popularity score”, which was an unclear measure, with an “usage” alternative, clearly defined in [7], that can be summarized as the percentage of games in which a particular champion appears.

This resulted in a tool, available online at <http://shiny.stlawu.edu:3838/LoLNOVA/>, which also provides the full dataset the authors used for their analysis. In this paper we are going to use the same dataset, which contains, in essence, the daily usage per champion across player tiers (normal to diamond), across regions (North America, Europe, Korea and all together) and also relates them to the free rotation schedule (whether they were free on that particular day, free on the previous one or two weeks or, else, not recently free). Data range has been restricted to League of Legends Season 5 (from November 11th 2014 to November 10th 2015) for the purpose of this analysis.

3. DATA ANALYSIS AND RESULT

3.1 Previous findings

As we are building from [7] it is worth to summarize Lee and Ramler’s findings before going further.

3.1.1 Impact of Free Rotation to Champion Usage

According to Lee and Ramler [7], there is a significant increase in average usage for free champions, especially in normal tier play (diminishing as it gets to higher ranked tiers). On the other hand, free rotation impacts usage for a longer period the higher the rank.

It’s also worth noting that the increase in usage is also related to Champion cost, with the most notable increase in the most expensive ones.

3.1.2 New and Updated Content

According to Lee and Ramler [7], the introduction of new champions and skins, and major and minor changes to Champions also impact Champion Usage. New Champions trigger high level of Usage that stabilizes over the following months and a similar phenomenon happens when existing Champions are reworked – although to a lesser extent. Minor balance tweaks, usually in regular patches, also result in a higher or lesser Usage for the affected Champions. The release of new skins, which are only aesthetic features, also impact Usage for the receiving Champions but the effect is not as large, as it mainly affects players that already owned the Champion and the effect is not persistent after a few days.

3.2 Special Events

The results that were summarized in the previous section show that new and updated content have relevant impact in usage. In this section we are moving the focus to special events and, for that purpose, the biggest available example from season 5 is the Bilgewater Event. This brings us to the first research question.

R.Q.1: Do special events have an impact in Champion Usage?

3.2.1 Bilgewater Event

The Bilgewater Event took place during three weeks between the 21st of July and the 10th of August 2015 and intertwined gameplay with lore. It impacted the game at various levels: first, it introduced a new game mode, called Black Market Brawlers, which won’t be analyzed in this paper but served as an interactive narrative to the lore update of certain champions. Second, it introduced an approximate ten thousand words story in three acts that aimed to provide background details to, mainly, four Champions: Twisted Fate (TF), Graves, Miss Fortune (MF) and Gangplank (GP)¹. Third, and most relevant to the current analysis, Miss Fortune and, especially, Gangplank were fully reworked; this means that their abilities and aesthetics were changed over the Bilgewater Event. In a somewhat controversial move from Riot, Gangplank was actually

¹ Available at <http://goo.gl/OxLiGg>.

disabled for a few days – leaving players unable to play him – to account for his death in the companion story.

On top of these, the free rotation during the two first weeks of the event was extended to accommodate both the eight usual Champions included plus the seven Bilgewater-related Champions, which included Fizz, Nami and Nautilus. Did the Bilgewater Event have a relevant impact in Champion usage? What about the MF and GP rework?

3.2.2 Results

3.2.2.1 Bilgewater Free Rotation Usage

First, we'll look at whether the impact of the Free Rotation is different for the seven Bilgewater Champions over the two weeks that the Bilgewater Event lasted compared to other regular Free Rotations for the same Champions. We are only considering Normal Tier Usage, as other tiers only allow owned Champions to be picked.

Figure 1. Bilgewater Free Rotation compared to regular Free Rotation for all Champions (season 5) compares, for all League of Legends Champions, the Free Rotation Usage during the Bilgewater Event (labeled as Bilgewater) on the related champions to all other Free Rotation Usage for all champions on all periods.

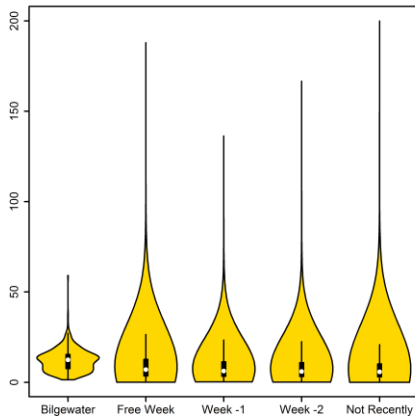


Figure 1. Bilgewater Free Rotation compared to regular Free Rotation for all Champions (season 5)

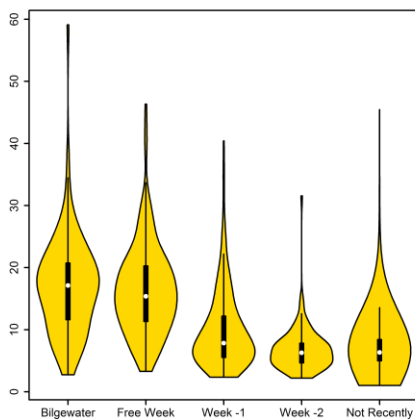


Figure 2. Bilgewater Free Rotation compared to regular Free Rotation for Bilgewater Champions only

As we can't assume variables to have a normal distribution and samples are not equal size, a Kruskal-Wallis one-way analysis of variance is run for the first case results in a significant difference found among buckets ($p < 0.001$). When running the corresponding Mann-Whitney U tests in order to identify which pairs differ from each other, the Bilgewater bucket is the one that appears to be significantly ($p < 0.001$) higher than the others in average.

As said, nevertheless, more than a hundred different Champions are available in League of Legends and, thus, the Free Week bucket contains many Champions whose popularity might be so low that hinder our results. To verify this, another Kruskal-Wallis test is run using only the seven Bilgewater Champions for the full season (Figure 2). Again, a significant difference is found ($p < 0.001$). This time, however, after running the Mann-Whitney U tests it is found that the Bilgewater Free Rotation Usage is no different from any other Free Rotation for the selected Champions (and both are statistically higher than when not in Free Rotation).

Therefore, as shown in [7], Champion Usage is relevantly higher when in Free Rotation but the same conclusion can't be reached about Special Events, that doesn't seem to have a relevant impact in Champion Usage overall.

3.2.2.2 Bilgewater Champions Detailed Usage

Even if the Bilgewater Event didn't drive Usage in general terms, it's worth taking a deeper look at Champions individually. Comparison between Bilgewater and other Free Rotations is provided in Figure 3.

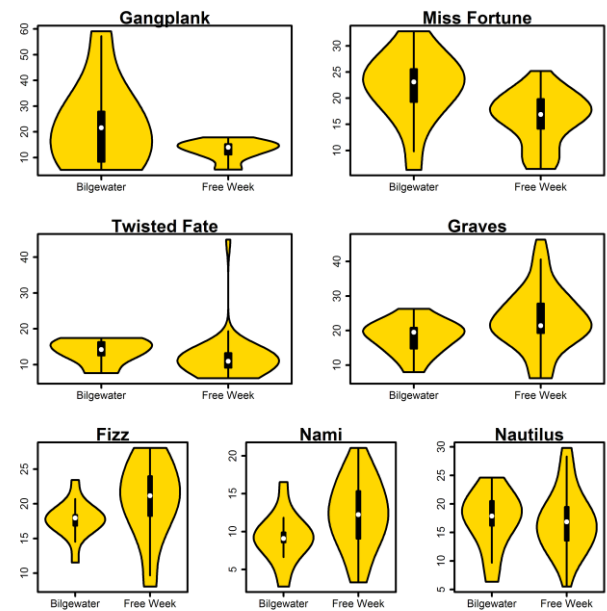


Figure 3. Bilgewater Free Rotation compared to regular Free Rotation by Bilgewater Champion

After a close inspection and the corresponding Mann-Whitney U tests, with its p values shown along with mean values in Table 1, a few interesting observations can be done.

- For some Champions, not only the Bilgewater effect doesn't raise usage but actually makes it significantly lower. With a confidence of 95% we can say that Fizz, Nami and Graves had a lower usage during Bilgewater than they did in other previous or later Free Rotations

during the same season. For Nautilus, on the other side, Usage kept virtually the same.

- Gangplank is a particular case as his Usage during the Bilgewater period (where he was one of the main characters) is quite irregular, as shown in Figure 4. Gangplank was reworked and that brought a peak in Usage over the first days, something that was also observed in [7], that raises the overall mean. However, the p value is higher than 0.05 and, therefore, we can't say that this difference in means is relevant. It's also interesting to note that GP was disabled during a few days at the beginning of August in a unique way: he was stated as dead [8]; every single player that logged in on those days noticed it. Upon reactivation (and an additional visual update) Usage peaked again in a similar fashion to a Free Rotation but, again, highly impacted by the introduction of new and updated content and not relevant in the current comparison.
- MF and TF are the only two Champions that actually show an increased Usage during the Bilgewater period that shows as relevant to the Mann-Whitney U test. That being said, however, there are two additional things that need to be taken into consideration. First, MF had a major rework during the Bilgewater event. Therefore, the increase in Usage can also be traced to that effect. TF was one of the event's protagonists but also received updated content (in this case, a new skin) so the increase in Usage cannot be assigned exclusively to the Special Event.

Table 1. Usage Means per Champion

Free Zone	GP	MF	TF	Grav.	Fizz	Nami	Naut.
Bilgewater	22.3	22.1	13.7	18.2	17.7	9.2	17.2
Free Rot.	12.8	16.3	11.7	22.9	20.4	12.3	16.9
p	0.190	0.002	0.011	0.049	0.028	0.019	0.525

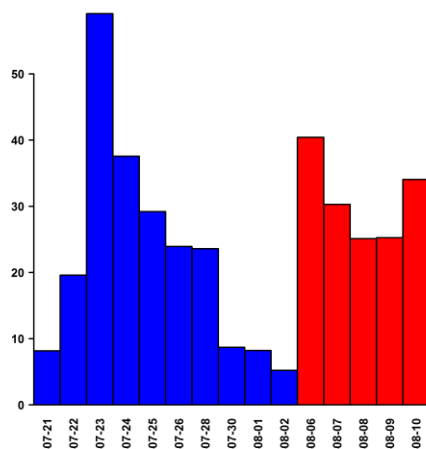


Figure 4. Gangplank Usage during the Bilgewater Event. Blue is for free rotation, Red for not free.

All in all, results show that new and updated content can bring an increase in Usage but, after looking at the biggest lore-related event of season 5, Special Events don't do that by themselves.

3.3 Event-driven motivation to read and learn

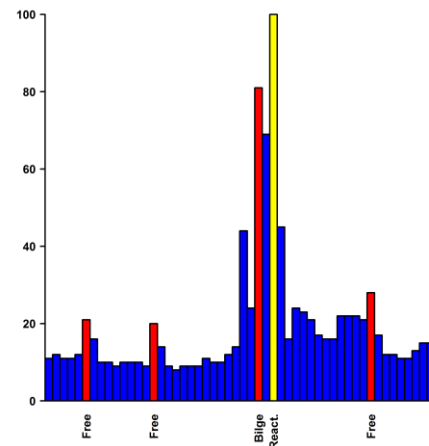
Even if Special Events don't drive Usage, Figure 4 showed an interesting insight: when changes in content or gameplay are made relevant and noticeable (such as GP story-driven death that disabled him for a few days), they might have an impact in user's (or player's) curiosity. That curiosity could turn into motivation to read about what is happening or to learn more about that particular Champion.

R.Q.2: Could events or videogame lore motivate to read and learn?

To that effect, we'll look specifically at GP, who had two main milestones during the Bilgewater Event, of which he was an integral part: he was fully reworked at the beginning and he was killed towards the end, with a new visual update being rolled over.

3.3.1 Interest over time

In order to measure how interest varies over time related to Gangplank, Google Trends will be used². Google Trends analyzes a percentage of Google web searches to figure out how many searches were done over a certain period of time. Duplicate searches by the same people and terms with low volume are excluded. Then, it assigns 100 to the week with the largest volume and the rest are assigned in relative values versus that volume. A search for Gangplank during 2015 yields the following results, shown in Figure 5.



3. The death and posterior reactivation of GP, with the visual update, brought the higher interest of all year, approximately ten times that of the previous non-free weeks and five times higher than the average of the posterior interest in the Champion.

3.3.2 Search Terms

Interest over time for Gangplank varies according to free rotations, events and reworks. But what do players search for? Table 2 shows the top 10 search terms over the full 2015 period and over July and August 2015 in particular, where the Bilgewater Event and GP rework took place.

Table 2. Gangplank Search Terms

Search Term	Year 2015 (Relative Freq.)	Jul-Aug 15 (Relative Freq.)	Type
Gangplank counter	100	80	Gameplay
Gangplank build	100	100	Gameplay
Gangplank lol	50	70	Generic
Gangplank guide	35	40	Gameplay
Gangplank top	30	NA	Gameplay
Gangplank s5	30	NA	Gameplay
Skin Gangplank	25	55	Generic
Captain Gangplank	20	50	Bilgewater
Gangplank rework	20	40	Bilgewater
Gangplank counters	15	NA	Gameplay
Gangplank dead	NA	35	Bilgewater
New Gangplank	NA	35	Bilgewater
League Gangplank	NA	45	Generic

Grouped as three main blocks, we identify:

1. A couple of terms are generic, thus resulting of little interest for the analysis.
2. Gameplay copes the top spots on the results. That basically means that most searches are centered on understanding how GP is played and built – and how it’s countered. “Build” is the term usually used to express how a Champion is itemized, what abilities are picked and in which particular order and, overall, how the Champion works.
3. The Bilgewater Event brings onto the table a few unique search terms that are specific to the event and some even make it to the top for the full year. “Captain Gangplank” is the skin released during the Bilgewater Event (so, we should also consider adding “Skin Gangplank” to the Bilgewater count). “Gangplank dead” at a relative index 35 also shows how that particular decision from Riot had a relevant impact on player’s interest. And last but not least, what’s interesting is to see how the “Gangplank rework” and “New Gangplank” show at the top spots in the Bilgewater period. A good number of players researched online in order to understand what changed and how to adapt GP to the new situation.

3.3.3 Discussion

According to Gee [9], “We always learn something”. As discussed in [10], “Gee argues that videogames are a semiotic domain, that learning to play a game (or genre of game) is to gain a new form of

literacy (one of many, as literacies are multiple), and that the study of games and learning is important to our understanding of how an individual comes to understand the *design grammar* of a semiotic domain generally”.

In [11], as cited in [10], Roberts et al. found that youth who spent the most time playing videogames actually spent more time reading than those who play less. Furthermore, in [12], Steinkuehler extends all previous work showing not only the complexity of digital and print textual practices involved in online gaming but also demonstrating that such practices are, when not higher, at least at level with the standards in reading, writing and technology.

“MMOs are new (albeit virtual) “third places” for informal sociability that are particularly well suited to the formation of bridging social capital” [13]. Although Steinkuehler and Williams are referring mainly to MMORPGs such as World of Warcraft when talking about MMOs, MOBAs such as League of Legends can be seen as a natural extension as they are also massive, multiplayer and online. “These communities set and determine what knowledge, skills, performances, values and dispositions are valued and therefore encouraged and which are not” [13]. So, when players search online, either broadly (such as Google) or specifically (such as in forums, specialized portals or even YouTube), what they are actually doing is understanding how the community thinks about, for example, Gangplank’s build or potential, and they are taking advantage of the huge amounts of collective intelligence [14] available online.

“Literacy, in the context of games, functions as a tool for solving problems” [10]. We can look at League of Legends as a complex system without a defined solution, an optimization problem, where a significant amount time is spent assessing how choices affect gameplay outcome. With limited gold earning during the game, the so-called build paths (basically, ways of building the desired items from the basic components to the final items) become essential. With limited experience and only one ability upgrade per level, the distribution of ability points can make the difference between winning and losing. With more than a hundred Champions to play and to play against, synergies and counters become also essential to understand in a never-ending network of abilities, items and stats. And the need to solve this vast problem (or, at least, to find a reasonably optimal or satisfying solution) arises from the same individual that acknowledges a gap between where he or she is at the moment and where he or she wants to be.

While it’s true that all this knowledge might not be required for the newest players to the game, as one progresses in League of Legends all available tools become increasingly relevant and important to succeed in the game. However, finding the information one needs within this “constellation of literacy artifacts” [12] is non-trivial. A simple search for “Gangplank” provides more than five million hits in Google and “Gangplank build” almost half a million. “Google is a big help, but only if one has a decent strategy for using it” [10].

4. CONCLUSION AND FUTURE WORK

In this article we have analyzed a Special Event (Bilgewater) that took place in League of Legends during summer of 2015, in the middle of the fifth season of the game. We followed to lines: first, we checked and assessed the impact of this Special Event in the game, particularly in Champion Usage. Results showed that, although there was an increase in the mean Usage for the main Bilgewater-related Champions (GP, MF and TF) this link was not found as statistically relevant. Both GP and MF received full reworks for the occasion (showing a common pattern with other reworks) and TF also received new content while being one of the

main characters of the Bilgewater story, so the effect is unclear (and that would make it the only Champion with a relevant increase during the free period out of seven). That being said, however, what we also found is a particular characteristic for Gangplank: the Champion was disabled following the events of the Bilgewater story and upon his return, the Usage peaked again as if he was in Free Week Rotation (but he wasn't). So, while lore and stories related to the game might not drive Usage, they might drive interest (or, at least, in some relevant events such as GP demise).

So, after noticing this behavior, in the second research question we analyzed whether the Special Event had an impact in player's interest about GP and what were the reasons of that interest. "From such a socio-cultural perspective, the most important point in understanding how games engage players in educational environments may be that good games engage players in multiple ways and the interplay between these different forms creates dynamic learning opportunities" [17] (as cited in [15]). And what was found was that during the Bilgewater event the searches in Google for Gangplank went noticeably up by a factor that ranged from five to ten versus a regular week, a figure much higher than the change in Usage. On top of that, most of the searches were about understanding and learning how to play the Champion (or how to play against it). Moreover, a few searches with relevant volume were observed that aimed to read about the "New Gangplank", the "Gangplank rework" or GP death. The single exercise of looking it up in Google, having to filter among hundreds of thousands of entries in forums, websites and other sources makes this a non-trivial research task for the player.

"Furthermore, with particular regard to the more modern game technologies such as multiplayer gaming, these interactions are becoming more complex, since multiplayer games can be played against real people anywhere in the world or against the computer, which means that the multiplayer games can be played at anytime, anywhere and by anyone, upgrading the complexity of human interactions and forming large communities in an expanded and more complex socio-cultural context" [16] (as cited in [15]).

If there is one relevant limitation to the current study is the number of events. A single event was analyzed (Bilgewater) because only one event happened in all 2015 or season 5. Lore events are rare and Bilgewater was the biggest experiment run by Riot that aimed to merge lore with gameplay. So, even if some of the findings could be promising, more evidence would need to be gathered from future events to fully validate the findings.

Searches are also driven by the need to assess the player's choice versus the community agreement on builds, by the need to express one's opinion about the changes and the rework on forums or other sites and even by the simple need of watching the latest skin or visual aspect of the Champion. All these aspects could be the base of further study, especially to understand how players received the event outside the game. Did they agree with the events of the story? Did they believe it? Was any fan-fiction or alternate story produced by players for the occasion? Did they like the Champions more after the event?

While lore and stories in League of Legends doesn't seem to directly influence gameplay, we have shown that they have the potential "to give rise to important digital literacy practices, that these digital literacy practices are cognitively complex [...] and that they can be seen as aligned with the acknowledged goals of education [...] rather than in competition with them" [10].

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